METR RESEARCH BRIEF

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every ~7 Months

50% task-completion time horizon: GPT-2 (~2 sec) → o3 (~1 hr 50 min)

March 2025

arXiv:2503.14499 | Kwa, West, Becker, et al.

A New Metric: 50% Task-Completion Time Horizon

1 Assemble Task Suite

170 diverse software/research tasks spanning 1 sec – 8 hrs

HCAST + RE-Bench + SWAA

2 Measure & Benchmark

Time skilled humans on each task; measure AI success rates

800+ human baselines, 2,529 hrs

3 Fit Logistic Model

Estimate task duration at which AI achieves 50% success rate

Human-grounded, comparable

Figure: Methodology Overview

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Source: Kwa et al., METR (2025)

BENCHMARK SUITE

170 Tasks Spanning 3 Seconds to 8 Hours

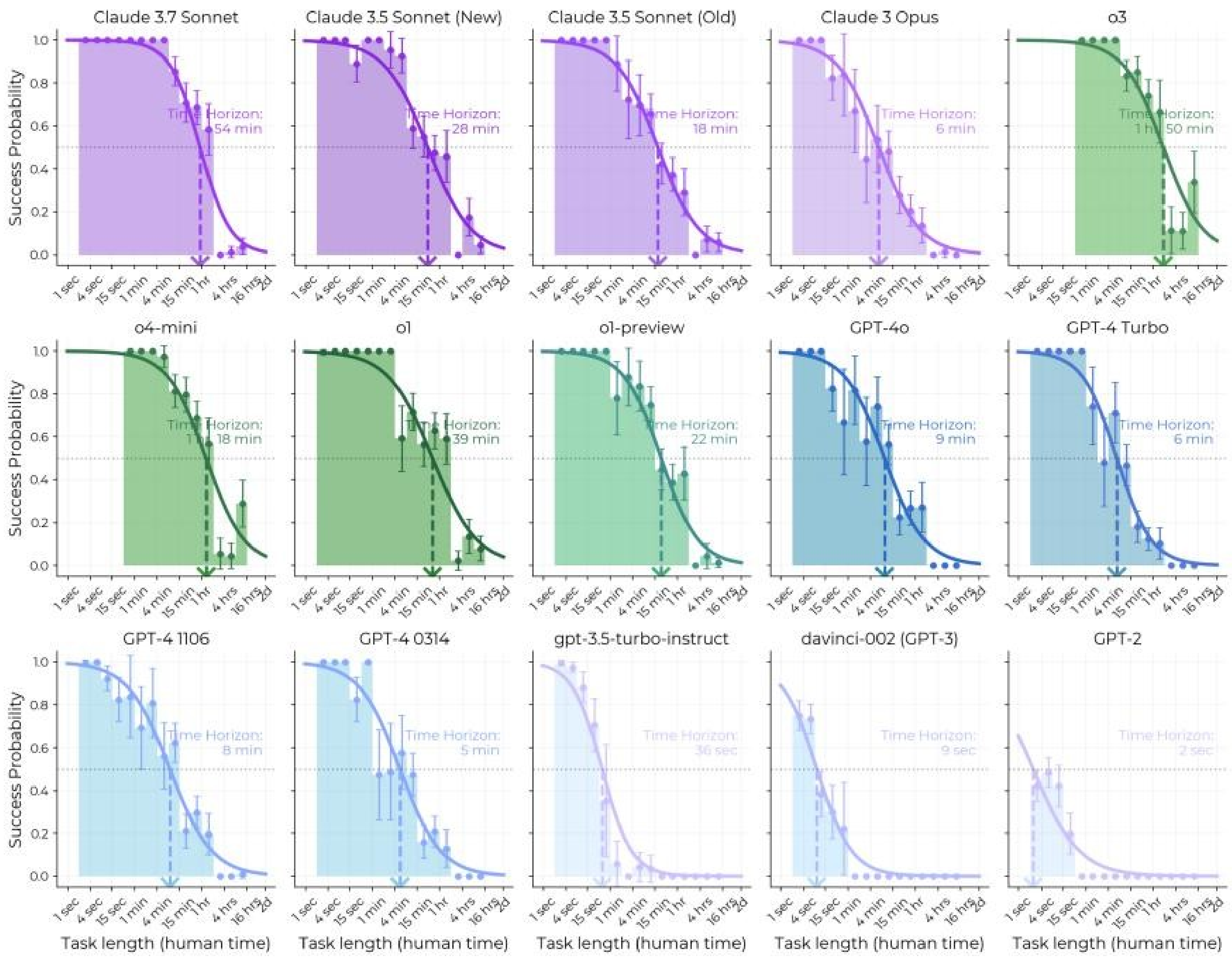
97 HCAST tasks (1 min – 30 hrs)	**7** RE-Bench tasks (all 8 hrs)	**66** SWAA tasks (1 sec – 30 sec)	**2,529** human hours collected (800+ baselines)

Example Tasks by Human Time

Task Family	Human Time	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?"
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 minutes	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30× speedup of a backtesting tool

Task sources: HCAST (diverse software), RE-Bench (ML research engineering), SWAA (short single-step actions)

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes (o3)



Key Statistics

Doubling time

~207 days

≈ 7 months

Fit quality

$R^2 = 0.97$

Over 6 years of releases

Acceleration signal

~20% faster

2023–2025 vs 2019–2025

Milestone Models

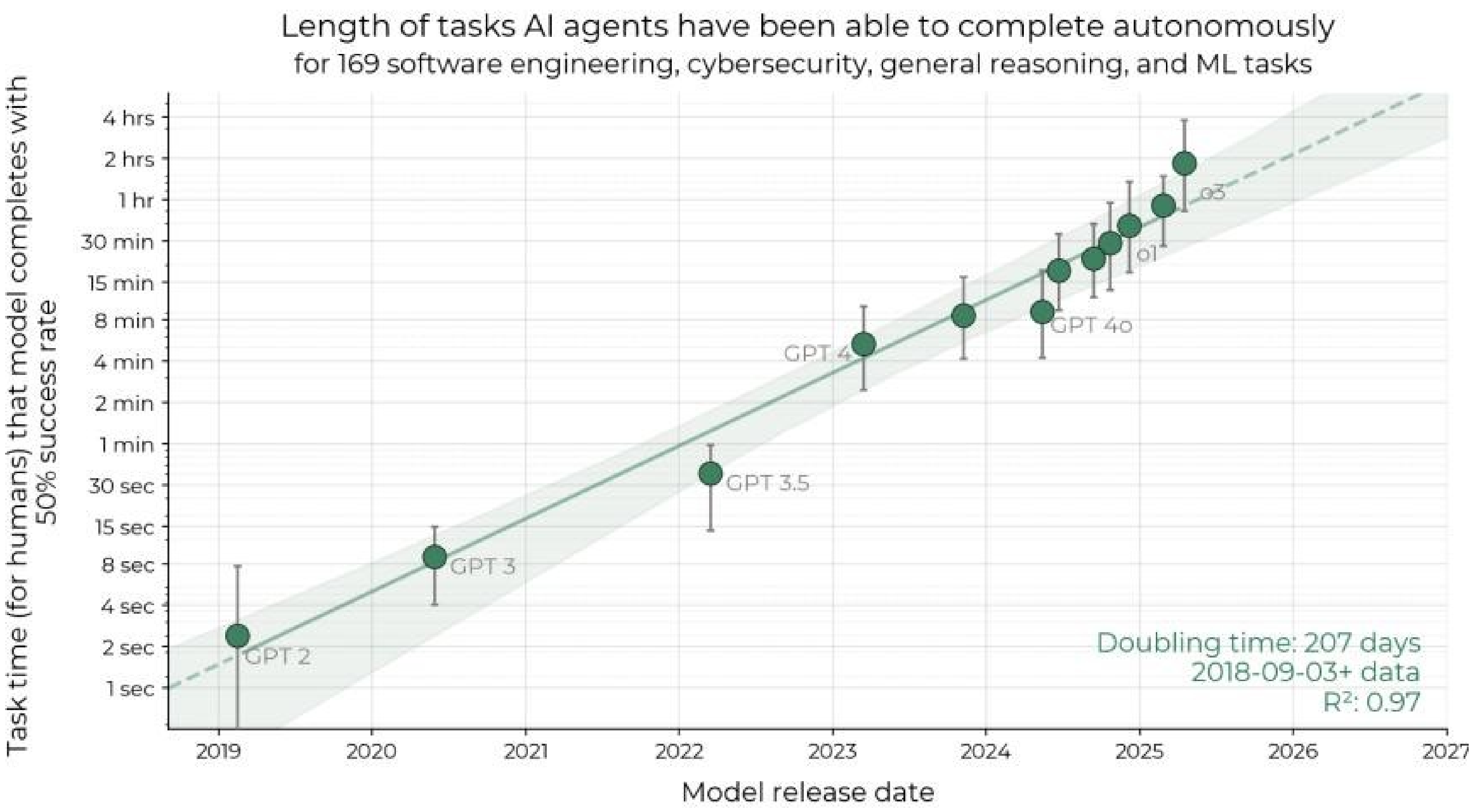
GPT-2 (2019): **~2 sec**

GPT-4 (2023): **~5 min**

o3 (2025): **~1 hr 50 min**

Source: Kwa et al., METR (2025)

Success Probability Drops Sharply as Task Length Increases



Model Comparison

Frontier (2025)

o3: ~1h 50m

Claude 3.7 Sonnet: ~54 min

Mid-tier (2024)

o1: ~39 min

Claude 3.5 Sonnet: ~28 min

GPT-4o: ~9 min

Baseline (2019)

GPT-2: ~2 sec

Key Insight

Every model's success rate declines with task length, but frontier models push the curve rightward by orders of magnitude.

Source: Kwa et al., METR (2025)

Reasoning, Tool Use, and Error Recovery Drive Growth

Improved Logical Reasoning

Models plan multi-step solutions more reliably, decomposing complex problems into tractable sub-tasks without losing coherence.

Key enabler for tasks >10 minutes

Better Tool Use

More effective use of code interpreters, terminals, file systems, and APIs to extend capabilities beyond pure text generation.

Critical for software engineering tasks

Reliability & Self-Awareness

Models increasingly know when they are uncertain, request clarification, and avoid hallucinated outputs that derail long tasks.

Reduces catastrophic failure rate

Error Recovery & Adaptation

Ability to detect own mistakes, backtrack, and try alternative approaches when initial strategies fail during extended execution.

Essential for tasks >30 minutes

Limitation: Performance is notably lower on less structured, "messier" tasks. External validity remains an open question.

Source: Kwa et al., METR (2025)

FORWARD PROJECTION

Extrapolation Predicts Month-Long Task Automation by 2028–2031

2019
GPT-2
~2 sec

2023
GPT-4
~5 min

2025
o3
~1h 50m

2028 – 2031
Month-long tasks
(167 work hours)

Projection

1 month

Time horizon by 2028–2031

≈ 167 work hours of human effort

Acceleration signal

~20% faster

2023–2025 vs 2019–2025 rate

Suggests possible trend acceleration

Implication

Automation of

Many month-long software projects

If naive exponential trend continues

Caveats & Uncertainties

- External validity: Tasks may not perfectly represent real-world intellectual labor distributions
- Trend changes: Exponential extrapolation is naive; hardware, algorithmic, or data bottlenecks may slow progress
- Unstructured tasks: Performance drops significantly on "messier" real-world problems not captured by benchmarks

Source: Kwa et al., METR (2025)

KEY TAKEAWAYS

Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1

Robust Metric

50% time horizon tracks AI autonomy with $R^2 = 0.97$ over 6 years

Human-grounded, intuitive, comparable across models

2

Doubling Every ~7 Months

Consistent exponential growth; possible acceleration since 2024

2023–2025 rate ~20% faster than 2019–2025 baseline

3

Current Frontier: ~2 Hours

o3 autonomously completes tasks requiring ~1 hr 50 min of human work

Up from 2 seconds for GPT-2 in 2019

4

External Validity Open Question

Tasks may not perfectly represent real-world intellectual labor

Supplementary experiments show little evidence of slower trends

5

Key Limitations: Unstructured Tasks & Generalization

Performance drops on "messier" tasks; uncertain generalization to real-world job distributions

Month-long automation projection (2028–2031) carries significant uncertainty

Safety governance must keep pace with capability growth.

Proactive evaluation, monitoring, and policy frameworks are essential.